

AI AssemblyTwin

Real-Time AI Digital Twin for Vehicle Assembly Manufacturing

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GitHub: github.com/jay-git00/DigitalTwin.ai

What It Does

AI AssemblyTwin is a production-grade digital twin of a 45-station vehicle assembly line. It combines a real-time SimPy simulation, four trained ML models, and a full-stack web application to deliver predictive quality and maintenance intelligence — before defects reach the customer. Unlike dashboards that replay historical data, AssemblyTwin runs a live simulation, streams telemetry at sub-100ms latency, and supports multi-stakeholder decision-making from the shop floor to the boardroom.

Quick Start

Requirements: Python 3.11+ · Node.js 20+

Terminal 1 — Backend:

```
cd backend
pip install -r requirements.txt
uvicorn main:app --reload --port 8000
```

Terminal 2 — Frontend:

```
cd frontend
npm install
npm run dev
```

Open: <http://localhost:3000>

Pages & Features

Page	URL	What You See
Live Floor	/	Factory map · Health gauge · ROI ticker · Before/After toggle · WHY?
Alerts	/alerts	Real-time alert feed · Intervention simulator · Approve / dismiss
Analytics	/analytics	4 tabs: Supervisor · Manager · Leadership · ESG / Sustainability
Defect Trace	/defect-trace	Causal defect chain from origin station → QC Gate
Maintenance	/maintenance	AI-predicted monthly maintenance calendar · Proactive cost comparison
Multi-Site	/multisite	3-plant enterprise overview: Chennai · Pune · Bangalore

ML Models

Model	Algorithm	Output
Anomaly Detector	Isolation Forest	<code>anomaly_score</code> ∈ [-1, 1]

Bottleneck Predictor	LSTM (NumPy)	bottleneck_prob per station $\in [0,1]$
Defect Predictor	Random Forest	defect_risk + feature importance
Sensor Imputer	Gaussian Process	imputed_value + σ (uncertainty)

API Reference

Endpoint	Description
WS /ws/live	Real-time station telemetry stream (every 3 sec)
GET /api/stations/status	All 45 stations: cycle_time, anomaly_score, defect_risk
GET /api/alerts	Active + historical alerts with severity
GET /api/system/health	Factory health score 0–100%
GET /api/explainability/{id}	SHAP-style feature importance for a station
GET /api/sparkline/{id}	Last 20 cycle-time readings
GET /api/maintenance/schedule	AI-predicted maintenance calendar + due dates
GET /api/multisite	3-plant enterprise network health
GET /api/esg	CO $\blacksquare$, steel, energy saved — ESG metrics
GET /api/causal/chain/{id}	Defect propagation chain: origin $\rightarrow$ QC Gate
GET /api/roi	Cumulative ROI stats

ROI Summary

Metric	Value
Deployment cost	■45 Lakh
Monthly savings	■18 Lakh
Payback period	~2.5 months
3-year net ROI	■6.1 Crore
Emergency repair multiplier	8.3 $\times$ vs proactive
New plant onboarding	4–6 weeks (vs 6+ months SCADA)

Team

- Jayanth — AI & System Architecture
- Abhinav — Industrial Integration & Telemetry
- Sagar — Frontend UX & Data Analytics